

CSA1717-ARTIFICIAL INTELLIGENCE PRACTICAL LAB EXPERIMENTS

FRUIT AND ITS COLOR USING BACKTRACKING

Aim:

To implement backtracking in Prolog to identify fruits and their colors.

Objective:

To demonstrate how Prolog searches through multiple possible solutions using backtracking.

Algorithm:

- 1. Start.**
- 2. Define fruit-color facts.**
- 3. Query a fruit or color.**
- 4. Prolog searches for a matching fact.**
- 5. If multiple matches exist, Prolog uses backtracking to find the next solution.**
- 6. Display all possible solutions.**
- 7. Stop.**

Flowchart: START

| v

Store Fruit-Color Facts

| v

Enter Fruit/Color

| v

Search Matching Fact

| v

Match Found?

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Yes No

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Display Fail Solution

| v

More Solutions?

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Yes No

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Backtrack STOP Prolog Program:

% Fruit and Color using Backtracking

```
fruit_color(apple, red). fruit_color(apple, green). fruit_color(banana, yellow).  
fruit_color(orange, orange). fruit_color(grapes, green). fruit_color(grapes, purple).  
fruit_color(mango, yellow). fruit_color(strawberry, red). Query
```

?- fruit_color(apple, Color). Output:

Color = red ; Color = green. Another query:

?- fruit_color(Fruit, green). Output:

Result:

Thus, fruit and color information is successfully retrieved using Prolog backtracking.